

# Sealed Experiment Record

Sealed 2026-10-05 · Signed by the K-Veritas server  
Session 34a907d4 · Data hash 1c16...d177

## Abstract

This record seals 4 runs (kveritas\_arm.py, sft.yaml, 0.1, sft.yaml) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-05 05:45 and 06:21 UTC. Reported results: phi 0.849638, mer 5.16667, mcs 0.865169, uwec 0.769663. 4 invocations in total, 0 failed. Produced with K-Veritas [1].

### What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

## 1 Setup

| Item               | Value                                                            |
|--------------------|------------------------------------------------------------------|
| Session            | 34a907d4-eef2-4113-8d4c-abdd4dd19473                             |
| Initialized        | 2026-10-05T05:45:09Z                                             |
| Machine            | 167ec0d6dc9a8170                                                 |
| Operating system   | linux / amd64                                                    |
| CPU                | Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores                  |
| Memory             | 31.0 GB                                                          |
| GPU                | NVIDIA GeForce RTX 5060 Ti, 16311 MiB                            |
| Environment, run 1 | 7327c8211955cb3512a0c35720459adafe8cc8a60a1483686d5330594cb422f1 |
| Environment, run 2 | 7327c8211955cb3512a0c35720459adafe8cc8a60a1483686d5330594cb422f1 |
| Environment, run 3 | 7327c8211955cb3512a0c35720459adafe8cc8a60a1483686d5330594cb422f1 |
| Environment, run 4 | 7327c8211955cb3512a0c35720459adafe8cc8a60a1483686d5330594cb422f1 |
| Disclosure         | open                                                             |
| Source code, run 1 | 1d15ecda1a4313a790720b974f078a41a406eb0dbe0797c833bae54ce2516ac3 |
| Source code, run 2 | 1d15ecda1a4313a790720b974f078a41a406eb0dbe0797c833bae54ce2516ac3 |
| Source code, run 3 | 1d15ecda1a4313a790720b974f078a41a406eb0dbe0797c833bae54ce2516ac3 |
| Source code, run 4 | 1d15ecda1a4313a790720b974f078a41a406eb0dbe0797c833bae54ce2516ac3 |

## 2 Runs

| # | Command                                        | Duration               | Exit | Anchored   |
|---|------------------------------------------------|------------------------|------|------------|
| 1 | python cookbooks/kveritas_arm.py sft /tmp/k... | 1.9 seconds            | 0    | at run end |
| 2 | sebeni distill -c cookbooks/kveritas/sft.yaml  | 55.1 seconds           | 0    | at run end |
| 3 | sebeni train -c cookbooks/kveritas/sft.yaml... | 33 minutes 22 secon... | 0    | at run end |
| 4 | sebeni eval -c cookbooks/kveritas/sft.yaml     | 2 minutes 10 seconds   | 0    | at run end |

4 invocations in total: 4 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

### 3 Results

#### Run 1: kveritas\_arm.py

No metrics reported.

#### Run 2: sft.yaml

No metrics reported.

#### Run 3: 0.1

| Metric                   | Final value | Step               | Source   |
|--------------------------|-------------|--------------------|----------|
| loss                     | 1.30145     | 500 (500 readings) | explicit |
| eval_loss                | 1.32598     | 500 (10 readings)  | explicit |
| eval_runtime             | 22.1618     | 500 (10 readings)  | explicit |
| eval_samples_per_second  | 16.244      | 500 (10 readings)  | explicit |
| eval_steps_per_second    | 2.031       | 500 (10 readings)  | explicit |
| eval_entropy             | 1.36176     | 500 (10 readings)  | explicit |
| eval_num_tokens          | 1.52793e+07 | 500 (10 readings)  | explicit |
| eval_mean_token_accuracy | 0.768744    | 500 (10 readings)  | explicit |
| epoch                    | 2.4642      | 500 (10 readings)  | explicit |

Seed committed: seed:42, output line 1, 2026-10-05T05:46:08Z.

#### Run 4: sft.yaml

| Metric   | Final value | Step | Source   |
|----------|-------------|------|----------|
| phi_bam  | 0.823658    | 0    | explicit |
| mer_bam  | 0           | 0    | explicit |
| mcs_bam  | 0           | 0    | explicit |
| uwec_bam | 1           | 0    | explicit |
| phi_bmq  | 0.752429    | 0    | explicit |
| mer_bmq  | 3           | 0    | explicit |
| mcs_bmq  | 0.833333    | 0    | explicit |
| uwec_bmq | 0.666667    | 0    | explicit |
| phi_boz  | 0.916823    | 0    | explicit |
| mer_boz  | 0           | 0    | explicit |
| mcs_boz  | 0           | 0    | explicit |
| uwec_boz | 1           | 0    | explicit |
| phi_dtm  | 0.809387    | 0    | explicit |
| mer_dtm  | 0.5         | 0    | explicit |
| mcs_dtm  | 0.333333    | 0    | explicit |
| uwec_dtm | 0.666667    | 0    | explicit |
| phi_ful  | 0.980054    | 0    | explicit |
| mer_ful  | 3           | 0    | explicit |
| mcs_ful  | 0.75        | 0    | explicit |
| uwec_ful | 0.625       | 0    | explicit |
| phi_mey  | 0.768726    | 0    | explicit |
| mer_mey  | 0           | 0    | explicit |
| mcs_mey  | 0           | 0    | explicit |
| uwec_mey | 0.5         | 0    | explicit |
| phi_mku  | 0.908601    | 0    | explicit |
| mer_mku  | 12          | 0    | explicit |
| mcs_mku  | 0.923077    | 0    | explicit |

| Metric   | Final value | Step | Source   |
|----------|-------------|------|----------|
| uwec_mku | 0.538462    | 0    | explicit |
| phi_kao  | 0.970238    | 0    | explicit |
| mer_kao  | 25.3333     | 0    | explicit |
| mcs_kao  | 0.974359    | 0    | explicit |
| uwec_kao | 1           | 0    | explicit |
| phi_myk  | 0.7953      | 0    | explicit |
| mer_myk  | 10          | 0    | explicit |
| mcs_myk  | 0.909091    | 0    | explicit |
| uwec_myk | 0.5         | 0    | explicit |
| phi_ses  | 0.836952    | 0    | explicit |
| mer_ses  | 0           | 0    | explicit |
| mcs_ses  | 0           | 0    | explicit |
| uwec_ses | 1           | 0    | explicit |
| phi_spp  | 0.853018    | 0    | explicit |
| mer_spp  | 6.66667     | 0    | explicit |
| mcs_spp  | 0.909091    | 0    | explicit |
| uwec_spp | 0.545455    | 0    | explicit |
| phi_snk  | 0.781673    | 0    | explicit |
| mer_snk  | 0.333333    | 0    | explicit |
| mcs_snk  | 0.333333    | 0    | explicit |
| uwec_snk | 0.666667    | 0    | explicit |
| phi_taq  | 0.84844     | 0    | explicit |
| mer_taq  | 0.5         | 0    | explicit |
| mcs_taq  | 0.5         | 0    | explicit |
| uwec_taq | 1           | 0    | explicit |
| phi      | 0.849638    | 0    | explicit |
| mer      | 5.16667     | 0    | explicit |
| mcs      | 0.865169    | 0    | explicit |
| uwec     | 0.769663    | 0    | explicit |

  

| Inline claim | Value    | Phase | Output line |
|--------------|----------|-------|-------------|
| phi          | 0.849638 | eval  | 64          |
| mer          | 5.16667  | eval  | 66          |
| mcs          | 0.865169 | eval  | 68          |
| uwec         | 0.769663 | eval  | 70          |

## 4 Execution evidence

### Coherence

Coherent (0.30): the telemetry channels move together, as one process.

### Telemetry

| Run | Samples | CPU time | Peak memory | Peak GPU |
|-----|---------|----------|-------------|----------|
| 1   | 16      | 2.5 s    | 0.78 GB     | 0%       |
| 2   | 546     | 55.8 s   | 1.44 GB     | 0%       |
| 3   | 7200    | 2015.8 s | 4.24 GB     | 98%      |
| 4   | 1291    | 131.4 s  | 3.49 GB     | 23%      |

## 5 Provenance

### Run 1: kveritas\_arm.py

Disclosure open. 135 tracked files, 2 snapshots. Every snapshot is bound into the signature.

| # | Snapshot  | Time     | Root             | Changed |
|---|-----------|----------|------------------|---------|
| 0 | run_start | 05:45:09 | 0a1f37f88004b036 | 134     |
| 1 | run_end   | 05:45:11 | 32cfabf37bf320ae | 1       |

Activity: 134 files read, 1 files written, 1 subprocesses. Full list in Appendix B.

### Run 2: sft.yaml

Disclosure open. 135 tracked files, 3 snapshots. Every snapshot is bound into the signature.

| # | Snapshot  | Time     | Root             | Changed |
|---|-----------|----------|------------------|---------|
| 0 | run_start | 05:45:12 | 32cfabf37bf320ae | 135     |
| 2 | run_end   | 05:46:06 | 32cfabf37bf320ae | 0       |

1 snapshot with no file change omitted here; full timeline in Appendix B.

Activity: 158 files read, 0 files written, 0 subprocesses. Full list in Appendix B.

### Run 3: 0.1

Disclosure open. 135 tracked files, 3 snapshots. Every snapshot is bound into the signature.

| # | Snapshot  | Time     | Root             | Changed |
|---|-----------|----------|------------------|---------|
| 0 | run_start | 05:46:07 | 32cfabf37bf320ae | 135     |
| 2 | run_end   | 06:19:30 | 32cfabf37bf320ae | 0       |

1 snapshot with no file change omitted here; full timeline in Appendix B.

Activity: 137 files read, 0 files written, 1 subprocesses. Full list in Appendix B.

### Run 4: sft.yaml

Disclosure open. 135 tracked files, 3 snapshots. Every snapshot is bound into the signature.

| # | Snapshot  | Time     | Root             | Changed |
|---|-----------|----------|------------------|---------|
| 0 | run_start | 06:19:31 | 32cfabf37bf320ae | 135     |
| 2 | run_end   | 06:21:41 | 32cfabf37bf320ae | 0       |

1 snapshot with no file change omitted here; full timeline in Appendix B.

Activity: 139 files read, 0 files written, 0 subprocesses. Full list in Appendix B.

## 6 Verification

| Field            | Value                                                            |
|------------------|------------------------------------------------------------------|
| Signer           | Signed by the K-Veritas server                                   |
| Signed at        | 2026-10-05T06:21:42.666471+00:00                                 |
| Algorithm        | RSA-PSS-SHA256, 4096-bit key                                     |
| Key fingerprint  | eeac81653cf44ac029c4069cf2a06d9f                                 |
| Data hash        | 1c1699a5bb61643823134d4b20b8cbf76989b4df4327e1bd9ed91816f2e2d177 |
| Code bundle hash | 199eb8b7c51638eae96fa0c03f3f5834f1b16c35debd164a5a4999b69388553a |

Run anchors: 4 invocations were anchored at the server when they ended.

**Checks a verifier runs**

1. The stored canonical data hashes to the data hash above, and the signature verifies over data\_hash:nonce:signed\_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

**References**

- [1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

# Appendix

## A Full metric series

Run 1: kveritas\_arm.py

Run 2: sft.yaml

| Phase   | Line | Time     | CPU time | Memory  | GPU |
|---------|------|----------|----------|---------|-----|
| distill | 1    | 05:45:13 | 2.3 s    | 0.78 GB |     |

Run 3: 0.1

| Metric | Value   | Step | Line | Source   |
|--------|---------|------|------|----------|
| loss   | 1.39278 | 1    | 12   | explicit |
| loss   | 1.34282 | 2    | 14   | explicit |
| loss   | 1.33672 | 3    | 16   | explicit |
| loss   | 1.35592 | 4    | 18   | explicit |
| loss   | 1.38709 | 5    | 20   | explicit |
| loss   | 1.4346  | 6    | 22   | explicit |
| loss   | 1.35665 | 7    | 24   | explicit |
| loss   | 1.36754 | 8    | 26   | explicit |
| loss   | 1.47553 | 9    | 28   | explicit |
| loss   | 1.37561 | 10   | 30   | explicit |
| loss   | 1.42687 | 11   | 32   | explicit |
| loss   | 1.42508 | 12   | 34   | explicit |
| loss   | 1.42589 | 13   | 36   | explicit |
| loss   | 1.37847 | 14   | 38   | explicit |
| loss   | 1.40036 | 15   | 40   | explicit |
| loss   | 1.36483 | 16   | 42   | explicit |
| loss   | 1.45462 | 17   | 44   | explicit |
| loss   | 1.32734 | 18   | 46   | explicit |
| loss   | 1.35012 | 19   | 48   | explicit |
| loss   | 1.36144 | 20   | 50   | explicit |
| loss   | 1.44746 | 21   | 52   | explicit |
| loss   | 1.38708 | 22   | 54   | explicit |
| loss   | 1.43341 | 23   | 56   | explicit |
| loss   | 1.3832  | 24   | 58   | explicit |
| loss   | 1.36105 | 25   | 60   | explicit |
| loss   | 1.3543  | 26   | 62   | explicit |
| loss   | 1.38662 | 27   | 64   | explicit |
| loss   | 1.39435 | 28   | 66   | explicit |
| loss   | 1.36375 | 29   | 68   | explicit |
| loss   | 1.4201  | 30   | 70   | explicit |
| loss   | 1.4119  | 31   | 72   | explicit |
| loss   | 1.35969 | 32   | 74   | explicit |
| loss   | 1.35094 | 33   | 76   | explicit |
| loss   | 1.33503 | 34   | 78   | explicit |
| loss   | 1.41574 | 35   | 80   | explicit |
| loss   | 1.34964 | 36   | 82   | explicit |
| loss   | 1.40211 | 37   | 84   | explicit |
| loss   | 1.42141 | 38   | 86   | explicit |

| Metric                   | Value    | Step | Line | Source   |
|--------------------------|----------|------|------|----------|
| loss                     | 1.43959  | 39   | 88   | explicit |
| loss                     | 1.47107  | 40   | 90   | explicit |
| loss                     | 1.35259  | 41   | 92   | explicit |
| loss                     | 1.38591  | 42   | 94   | explicit |
| loss                     | 1.35482  | 43   | 96   | explicit |
| loss                     | 1.31499  | 44   | 98   | explicit |
| loss                     | 1.37929  | 45   | 100  | explicit |
| loss                     | 1.37297  | 46   | 102  | explicit |
| loss                     | 1.34736  | 47   | 104  | explicit |
| loss                     | 1.43037  | 48   | 106  | explicit |
| loss                     | 1.40051  | 49   | 108  | explicit |
| loss                     | 1.37353  | 50   | 110  | explicit |
| eval_loss                | 1.37382  | 50   | 113  | explicit |
| eval_runtime             | 22.1527  | 50   | 114  | explicit |
| eval_samples_per_second  | 16.251   | 50   | 115  | explicit |
| eval_steps_per_second    | 2.031    | 50   | 116  | explicit |
| eval_entropy             | 1.42372  | 50   | 117  | explicit |
| eval_num_tokens          | 1.52e+06 | 50   | 118  | explicit |
| eval_mean_token_accuracy | 0.759813 | 50   | 119  | explicit |
| epoch                    | 0.246914 | 50   | 120  | explicit |
| loss                     | 1.41456  | 51   | 121  | explicit |
| loss                     | 1.40333  | 52   | 123  | explicit |
| loss                     | 1.36886  | 53   | 125  | explicit |
| loss                     | 1.37114  | 54   | 127  | explicit |
| loss                     | 1.39942  | 55   | 129  | explicit |
| loss                     | 1.39832  | 56   | 131  | explicit |
| loss                     | 1.43754  | 57   | 133  | explicit |
| loss                     | 1.4311   | 58   | 135  | explicit |
| loss                     | 1.37561  | 59   | 137  | explicit |
| loss                     | 1.42916  | 60   | 139  | explicit |
| loss                     | 1.45878  | 61   | 141  | explicit |
| loss                     | 1.40727  | 62   | 143  | explicit |
| loss                     | 1.29612  | 63   | 145  | explicit |
| loss                     | 1.34043  | 64   | 147  | explicit |
| loss                     | 1.40594  | 65   | 149  | explicit |
| loss                     | 1.36517  | 66   | 151  | explicit |
| loss                     | 1.39091  | 67   | 153  | explicit |
| loss                     | 1.32914  | 68   | 155  | explicit |
| loss                     | 1.43025  | 69   | 157  | explicit |
| loss                     | 1.35024  | 70   | 159  | explicit |
| loss                     | 1.4613   | 71   | 161  | explicit |
| loss                     | 1.38618  | 72   | 163  | explicit |
| loss                     | 1.42491  | 73   | 165  | explicit |
| loss                     | 1.37567  | 74   | 167  | explicit |
| loss                     | 1.3864   | 75   | 169  | explicit |
| loss                     | 1.41291  | 76   | 171  | explicit |
| loss                     | 1.37218  | 77   | 173  | explicit |
| loss                     | 1.33754  | 78   | 175  | explicit |
| loss                     | 1.32374  | 79   | 177  | explicit |
| loss                     | 1.3622   | 80   | 179  | explicit |

| Metric                   | Value       | Step | Line | Source   |
|--------------------------|-------------|------|------|----------|
| loss                     | 1.35634     | 81   | 181  | explicit |
| loss                     | 1.35744     | 82   | 183  | explicit |
| loss                     | 1.41732     | 83   | 185  | explicit |
| loss                     | 1.351       | 84   | 187  | explicit |
| loss                     | 1.45841     | 85   | 189  | explicit |
| loss                     | 1.37323     | 86   | 191  | explicit |
| loss                     | 1.35144     | 87   | 193  | explicit |
| loss                     | 1.35393     | 88   | 195  | explicit |
| loss                     | 1.33726     | 89   | 197  | explicit |
| loss                     | 1.36972     | 90   | 199  | explicit |
| loss                     | 1.3565      | 91   | 201  | explicit |
| loss                     | 1.37068     | 92   | 203  | explicit |
| loss                     | 1.29712     | 93   | 205  | explicit |
| loss                     | 1.40036     | 94   | 207  | explicit |
| loss                     | 1.34019     | 95   | 209  | explicit |
| loss                     | 1.39738     | 96   | 211  | explicit |
| loss                     | 1.35216     | 97   | 213  | explicit |
| loss                     | 1.33963     | 98   | 215  | explicit |
| loss                     | 1.3494      | 99   | 217  | explicit |
| loss                     | 1.43184     | 100  | 219  | explicit |
| eval_loss                | 1.35926     | 100  | 222  | explicit |
| eval_runtime             | 22.1444     | 100  | 223  | explicit |
| eval_samples_per_second  | 16.257      | 100  | 224  | explicit |
| eval_steps_per_second    | 2.032       | 100  | 225  | explicit |
| eval_entropy             | 1.40768     | 100  | 226  | explicit |
| eval_num_tokens          | 3.04258e+06 | 100  | 227  | explicit |
| eval_mean_token_accuracy | 0.762127    | 100  | 228  | explicit |
| epoch                    | 0.493827    | 100  | 229  | explicit |
| loss                     | 1.29142     | 101  | 230  | explicit |
| loss                     | 1.37295     | 102  | 232  | explicit |
| loss                     | 1.3981      | 103  | 234  | explicit |
| loss                     | 1.37855     | 104  | 236  | explicit |
| loss                     | 1.36607     | 105  | 238  | explicit |
| loss                     | 1.39409     | 106  | 240  | explicit |
| loss                     | 1.37007     | 107  | 242  | explicit |
| loss                     | 1.39397     | 108  | 244  | explicit |
| loss                     | 1.30756     | 109  | 246  | explicit |
| loss                     | 1.425       | 110  | 248  | explicit |
| loss                     | 1.42531     | 111  | 250  | explicit |
| loss                     | 1.30635     | 112  | 252  | explicit |
| loss                     | 1.37012     | 113  | 254  | explicit |
| loss                     | 1.3467      | 114  | 256  | explicit |
| loss                     | 1.29614     | 115  | 258  | explicit |
| loss                     | 1.36499     | 116  | 260  | explicit |
| loss                     | 1.33962     | 117  | 262  | explicit |
| loss                     | 1.39185     | 118  | 264  | explicit |
| loss                     | 1.42665     | 119  | 266  | explicit |
| loss                     | 1.34853     | 120  | 268  | explicit |
| loss                     | 1.33317     | 121  | 270  | explicit |
| loss                     | 1.42059     | 122  | 272  | explicit |

| Metric                   | Value       | Step | Line | Source   |
|--------------------------|-------------|------|------|----------|
| loss                     | 1.34227     | 123  | 274  | explicit |
| loss                     | 1.38162     | 124  | 276  | explicit |
| loss                     | 1.42793     | 125  | 278  | explicit |
| loss                     | 1.34301     | 126  | 280  | explicit |
| loss                     | 1.33878     | 127  | 282  | explicit |
| loss                     | 1.40758     | 128  | 284  | explicit |
| loss                     | 1.28588     | 129  | 286  | explicit |
| loss                     | 1.35111     | 130  | 288  | explicit |
| loss                     | 1.29891     | 131  | 290  | explicit |
| loss                     | 1.4181      | 132  | 292  | explicit |
| loss                     | 1.36107     | 133  | 294  | explicit |
| loss                     | 1.29916     | 134  | 296  | explicit |
| loss                     | 1.39275     | 135  | 298  | explicit |
| loss                     | 1.37913     | 136  | 300  | explicit |
| loss                     | 1.28747     | 137  | 302  | explicit |
| loss                     | 1.28773     | 138  | 304  | explicit |
| loss                     | 1.32162     | 139  | 306  | explicit |
| loss                     | 1.32498     | 140  | 308  | explicit |
| loss                     | 1.39778     | 141  | 310  | explicit |
| loss                     | 1.33427     | 142  | 312  | explicit |
| loss                     | 1.40598     | 143  | 314  | explicit |
| loss                     | 1.34164     | 144  | 316  | explicit |
| loss                     | 1.33798     | 145  | 318  | explicit |
| loss                     | 1.31945     | 146  | 320  | explicit |
| loss                     | 1.43383     | 147  | 322  | explicit |
| loss                     | 1.26744     | 148  | 324  | explicit |
| loss                     | 1.42468     | 149  | 326  | explicit |
| loss                     | 1.38732     | 150  | 328  | explicit |
| eval_loss                | 1.3459      | 150  | 331  | explicit |
| eval_runtime             | 22.1441     | 150  | 332  | explicit |
| eval_samples_per_second  | 16.257      | 150  | 333  | explicit |
| eval_steps_per_second    | 2.032       | 150  | 334  | explicit |
| eval_entropy             | 1.38743     | 150  | 335  | explicit |
| eval_num_tokens          | 4.58232e+06 | 150  | 336  | explicit |
| eval_mean_token_accuracy | 0.765185    | 150  | 337  | explicit |
| epoch                    | 0.740741    | 150  | 338  | explicit |
| loss                     | 1.29644     | 151  | 339  | explicit |
| loss                     | 1.36772     | 152  | 341  | explicit |
| loss                     | 1.36395     | 153  | 343  | explicit |
| loss                     | 1.30383     | 154  | 345  | explicit |
| loss                     | 1.311       | 155  | 347  | explicit |
| loss                     | 1.36507     | 156  | 349  | explicit |
| loss                     | 1.30798     | 157  | 351  | explicit |
| loss                     | 1.32526     | 158  | 353  | explicit |
| loss                     | 1.37554     | 159  | 355  | explicit |
| loss                     | 1.31997     | 160  | 357  | explicit |
| loss                     | 1.36647     | 161  | 359  | explicit |
| loss                     | 1.26231     | 162  | 361  | explicit |
| loss                     | 1.35915     | 163  | 363  | explicit |
| loss                     | 1.30604     | 164  | 365  | explicit |

| Metric                   | Value      | Step | Line | Source   |
|--------------------------|------------|------|------|----------|
| loss                     | 1.34954    | 165  | 367  | explicit |
| loss                     | 1.26435    | 166  | 369  | explicit |
| loss                     | 1.33937    | 167  | 371  | explicit |
| loss                     | 1.29272    | 168  | 373  | explicit |
| loss                     | 1.373      | 169  | 375  | explicit |
| loss                     | 1.32867    | 170  | 377  | explicit |
| loss                     | 1.25283    | 171  | 379  | explicit |
| loss                     | 1.33817    | 172  | 381  | explicit |
| loss                     | 1.40647    | 173  | 383  | explicit |
| loss                     | 1.3275     | 174  | 385  | explicit |
| loss                     | 1.3271     | 175  | 387  | explicit |
| loss                     | 1.34114    | 176  | 389  | explicit |
| loss                     | 1.32163    | 177  | 391  | explicit |
| loss                     | 1.38107    | 178  | 393  | explicit |
| loss                     | 1.39881    | 179  | 395  | explicit |
| loss                     | 1.34971    | 180  | 397  | explicit |
| loss                     | 1.5067     | 181  | 399  | explicit |
| loss                     | 1.32687    | 182  | 401  | explicit |
| loss                     | 1.30843    | 183  | 403  | explicit |
| loss                     | 1.42199    | 184  | 405  | explicit |
| loss                     | 1.39336    | 185  | 407  | explicit |
| loss                     | 1.36007    | 186  | 409  | explicit |
| loss                     | 1.31869    | 187  | 411  | explicit |
| loss                     | 1.3132     | 188  | 413  | explicit |
| loss                     | 1.42414    | 189  | 415  | explicit |
| loss                     | 1.35927    | 190  | 417  | explicit |
| loss                     | 1.33199    | 191  | 419  | explicit |
| loss                     | 1.28801    | 192  | 421  | explicit |
| loss                     | 1.26575    | 193  | 423  | explicit |
| loss                     | 1.32298    | 194  | 425  | explicit |
| loss                     | 1.32439    | 195  | 427  | explicit |
| loss                     | 1.33712    | 196  | 429  | explicit |
| loss                     | 1.3148     | 197  | 431  | explicit |
| loss                     | 1.36716    | 198  | 433  | explicit |
| loss                     | 1.29242    | 199  | 435  | explicit |
| loss                     | 1.31407    | 200  | 437  | explicit |
| eval_loss                | 1.33344    | 200  | 440  | explicit |
| eval_runtime             | 22.1556    | 200  | 441  | explicit |
| eval_samples_per_second  | 16.249     | 200  | 442  | explicit |
| eval_steps_per_second    | 2.031      | 200  | 443  | explicit |
| eval_entropy             | 1.36704    | 200  | 444  | explicit |
| eval_num_tokens          | 6.1207e+06 | 200  | 445  | explicit |
| eval_mean_token_accuracy | 0.768469   | 200  | 446  | explicit |
| epoch                    | 0.987654   | 200  | 447  | explicit |
| loss                     | 1.36787    | 201  | 448  | explicit |
| loss                     | 1.38235    | 202  | 450  | explicit |
| loss                     | 1.33381    | 203  | 452  | explicit |
| loss                     | 1.32756    | 204  | 454  | explicit |
| loss                     | 1.29174    | 205  | 456  | explicit |
| loss                     | 1.3537     | 206  | 458  | explicit |

| Metric                  | Value       | Step | Line | Source   |
|-------------------------|-------------|------|------|----------|
| loss                    | 1.29517     | 207  | 460  | explicit |
| loss                    | 1.30755     | 208  | 462  | explicit |
| loss                    | 1.31067     | 209  | 464  | explicit |
| loss                    | 1.32471     | 210  | 466  | explicit |
| loss                    | 1.39756     | 211  | 468  | explicit |
| loss                    | 1.37972     | 212  | 470  | explicit |
| loss                    | 1.29603     | 213  | 472  | explicit |
| loss                    | 1.31782     | 214  | 474  | explicit |
| loss                    | 1.35864     | 215  | 476  | explicit |
| loss                    | 1.3377      | 216  | 478  | explicit |
| loss                    | 1.30501     | 217  | 480  | explicit |
| loss                    | 1.32363     | 218  | 482  | explicit |
| loss                    | 1.34265     | 219  | 484  | explicit |
| loss                    | 1.34159     | 220  | 486  | explicit |
| loss                    | 1.31953     | 221  | 488  | explicit |
| loss                    | 1.34514     | 222  | 490  | explicit |
| loss                    | 1.33448     | 223  | 492  | explicit |
| loss                    | 1.37717     | 224  | 494  | explicit |
| loss                    | 1.28178     | 225  | 496  | explicit |
| loss                    | 1.36373     | 226  | 498  | explicit |
| loss                    | 1.30058     | 227  | 500  | explicit |
| loss                    | 1.32666     | 228  | 502  | explicit |
| loss                    | 1.37451     | 229  | 504  | explicit |
| loss                    | 1.34921     | 230  | 506  | explicit |
| loss                    | 1.34574     | 231  | 508  | explicit |
| loss                    | 1.28461     | 232  | 510  | explicit |
| loss                    | 1.26671     | 233  | 512  | explicit |
| loss                    | 1.32022     | 234  | 514  | explicit |
| loss                    | 1.29679     | 235  | 516  | explicit |
| loss                    | 1.3414      | 236  | 518  | explicit |
| loss                    | 1.33436     | 237  | 520  | explicit |
| loss                    | 1.33213     | 238  | 522  | explicit |
| loss                    | 1.34863     | 239  | 524  | explicit |
| loss                    | 1.34384     | 240  | 526  | explicit |
| loss                    | 1.3049      | 241  | 528  | explicit |
| loss                    | 1.29054     | 242  | 530  | explicit |
| loss                    | 1.2906      | 243  | 532  | explicit |
| loss                    | 1.27103     | 244  | 534  | explicit |
| loss                    | 1.33867     | 245  | 536  | explicit |
| loss                    | 1.34122     | 246  | 538  | explicit |
| loss                    | 1.43502     | 247  | 540  | explicit |
| loss                    | 1.36202     | 248  | 542  | explicit |
| loss                    | 1.40944     | 249  | 544  | explicit |
| loss                    | 1.34434     | 250  | 546  | explicit |
| eval_loss               | 1.3245      | 250  | 549  | explicit |
| eval_runtime            | 22.1526     | 250  | 550  | explicit |
| eval_samples_per_second | 16.251      | 250  | 551  | explicit |
| eval_steps_per_second   | 2.031       | 250  | 552  | explicit |
| eval_entropy            | 1.35875     | 250  | 553  | explicit |
| eval_num_tokens         | 7.63496e+06 | 250  | 554  | explicit |

| Metric                   | Value    | Step | Line | Source   |
|--------------------------|----------|------|------|----------|
| eval_mean_token_accuracy | 0.769471 | 250  | 555  | explicit |
| epoch                    | 1.2321   | 250  | 556  | explicit |
| loss                     | 1.23578  | 251  | 557  | explicit |
| loss                     | 1.28456  | 252  | 559  | explicit |
| loss                     | 1.34599  | 253  | 561  | explicit |
| loss                     | 1.39164  | 254  | 563  | explicit |
| loss                     | 1.3254   | 255  | 565  | explicit |
| loss                     | 1.27737  | 256  | 567  | explicit |
| loss                     | 1.30978  | 257  | 569  | explicit |
| loss                     | 1.24062  | 258  | 571  | explicit |
| loss                     | 1.35496  | 259  | 573  | explicit |
| loss                     | 1.34562  | 260  | 575  | explicit |
| loss                     | 1.35314  | 261  | 577  | explicit |
| loss                     | 1.44312  | 262  | 579  | explicit |
| loss                     | 1.37492  | 263  | 581  | explicit |
| loss                     | 1.28058  | 264  | 583  | explicit |
| loss                     | 1.33419  | 265  | 585  | explicit |
| loss                     | 1.31439  | 266  | 587  | explicit |
| loss                     | 1.33494  | 267  | 589  | explicit |
| loss                     | 1.3608   | 268  | 591  | explicit |
| loss                     | 1.31709  | 269  | 593  | explicit |
| loss                     | 1.29339  | 270  | 595  | explicit |
| loss                     | 1.33215  | 271  | 597  | explicit |
| loss                     | 1.36952  | 272  | 599  | explicit |
| loss                     | 1.45321  | 273  | 601  | explicit |
| loss                     | 1.35743  | 274  | 603  | explicit |
| loss                     | 1.34236  | 275  | 605  | explicit |
| loss                     | 1.29353  | 276  | 607  | explicit |
| loss                     | 1.28857  | 277  | 609  | explicit |
| loss                     | 1.3751   | 278  | 611  | explicit |
| loss                     | 1.37506  | 279  | 613  | explicit |
| loss                     | 1.38932  | 280  | 615  | explicit |
| loss                     | 1.29887  | 281  | 617  | explicit |
| loss                     | 1.33664  | 282  | 619  | explicit |
| loss                     | 1.34378  | 283  | 621  | explicit |
| loss                     | 1.29066  | 284  | 623  | explicit |
| loss                     | 1.3286   | 285  | 625  | explicit |
| loss                     | 1.38529  | 286  | 627  | explicit |
| loss                     | 1.31225  | 287  | 629  | explicit |
| loss                     | 1.3058   | 288  | 631  | explicit |
| loss                     | 1.30172  | 289  | 633  | explicit |
| loss                     | 1.29329  | 290  | 635  | explicit |
| loss                     | 1.28096  | 291  | 637  | explicit |
| loss                     | 1.29599  | 292  | 639  | explicit |
| loss                     | 1.38126  | 293  | 641  | explicit |
| loss                     | 1.28121  | 294  | 643  | explicit |
| loss                     | 1.34268  | 295  | 645  | explicit |
| loss                     | 1.3268   | 296  | 647  | explicit |
| loss                     | 1.35637  | 297  | 649  | explicit |
| loss                     | 1.3101   | 298  | 651  | explicit |

| Metric                   | Value       | Step | Line | Source   |
|--------------------------|-------------|------|------|----------|
| loss                     | 1.34361     | 299  | 653  | explicit |
| loss                     | 1.32494     | 300  | 655  | explicit |
| eval_loss                | 1.3255      | 300  | 658  | explicit |
| eval_runtime             | 22.1544     | 300  | 659  | explicit |
| eval_samples_per_second  | 16.25       | 300  | 660  | explicit |
| eval_steps_per_second    | 2.031       | 300  | 661  | explicit |
| eval_entropy             | 1.35931     | 300  | 662  | explicit |
| eval_num_tokens          | 9.16232e+06 | 300  | 663  | explicit |
| eval_mean_token_accuracy | 0.768884    | 300  | 664  | explicit |
| epoch                    | 1.47901     | 300  | 665  | explicit |
| loss                     | 1.3408      | 301  | 666  | explicit |
| loss                     | 1.22427     | 302  | 668  | explicit |
| loss                     | 1.27546     | 303  | 670  | explicit |
| loss                     | 1.27206     | 304  | 672  | explicit |
| loss                     | 1.40892     | 305  | 674  | explicit |
| loss                     | 1.25484     | 306  | 676  | explicit |
| loss                     | 1.32378     | 307  | 678  | explicit |
| loss                     | 1.31526     | 308  | 680  | explicit |
| loss                     | 1.38356     | 309  | 682  | explicit |
| loss                     | 1.25132     | 310  | 684  | explicit |
| loss                     | 1.3922      | 311  | 686  | explicit |
| loss                     | 1.27507     | 312  | 688  | explicit |
| loss                     | 1.41128     | 313  | 690  | explicit |
| loss                     | 1.30766     | 314  | 692  | explicit |
| loss                     | 1.3266      | 315  | 694  | explicit |
| loss                     | 1.31406     | 316  | 696  | explicit |
| loss                     | 1.32828     | 317  | 698  | explicit |
| loss                     | 1.28077     | 318  | 700  | explicit |
| loss                     | 1.33337     | 319  | 702  | explicit |
| loss                     | 1.32923     | 320  | 704  | explicit |
| loss                     | 1.29226     | 321  | 706  | explicit |
| loss                     | 1.38731     | 322  | 708  | explicit |
| loss                     | 1.32184     | 323  | 710  | explicit |
| loss                     | 1.33433     | 324  | 712  | explicit |
| loss                     | 1.3351      | 325  | 714  | explicit |
| loss                     | 1.29673     | 326  | 716  | explicit |
| loss                     | 1.30177     | 327  | 718  | explicit |
| loss                     | 1.32526     | 328  | 720  | explicit |
| loss                     | 1.34112     | 329  | 722  | explicit |
| loss                     | 1.40346     | 330  | 724  | explicit |
| loss                     | 1.27561     | 331  | 726  | explicit |
| loss                     | 1.35517     | 332  | 728  | explicit |
| loss                     | 1.33975     | 333  | 730  | explicit |
| loss                     | 1.42703     | 334  | 732  | explicit |
| loss                     | 1.30568     | 335  | 734  | explicit |
| loss                     | 1.31035     | 336  | 736  | explicit |
| loss                     | 1.40609     | 337  | 738  | explicit |
| loss                     | 1.39175     | 338  | 740  | explicit |
| loss                     | 1.36727     | 339  | 742  | explicit |
| loss                     | 1.33724     | 340  | 744  | explicit |

| Metric                   | Value       | Step | Line | Source   |
|--------------------------|-------------|------|------|----------|
| loss                     | 1.39438     | 341  | 746  | explicit |
| loss                     | 1.40171     | 342  | 748  | explicit |
| loss                     | 1.28491     | 343  | 750  | explicit |
| loss                     | 1.39383     | 344  | 752  | explicit |
| loss                     | 1.40034     | 345  | 754  | explicit |
| loss                     | 1.30956     | 346  | 756  | explicit |
| loss                     | 1.46544     | 347  | 758  | explicit |
| loss                     | 1.34592     | 348  | 760  | explicit |
| loss                     | 1.22207     | 349  | 762  | explicit |
| loss                     | 1.25086     | 350  | 764  | explicit |
| eval_loss                | 1.32432     | 350  | 767  | explicit |
| eval_runtime             | 22.1474     | 350  | 768  | explicit |
| eval_samples_per_second  | 16.255      | 350  | 769  | explicit |
| eval_steps_per_second    | 2.032       | 350  | 770  | explicit |
| eval_entropy             | 1.35956     | 350  | 771  | explicit |
| eval_num_tokens          | 1.06909e+07 | 350  | 772  | explicit |
| eval_mean_token_accuracy | 0.769438    | 350  | 773  | explicit |
| epoch                    | 1.72593     | 350  | 774  | explicit |
| loss                     | 1.29434     | 351  | 775  | explicit |
| loss                     | 1.39847     | 352  | 777  | explicit |
| loss                     | 1.33158     | 353  | 779  | explicit |
| loss                     | 1.32833     | 354  | 781  | explicit |
| loss                     | 1.32493     | 355  | 783  | explicit |
| loss                     | 1.36864     | 356  | 785  | explicit |
| loss                     | 1.33101     | 357  | 787  | explicit |
| loss                     | 1.38432     | 358  | 789  | explicit |
| loss                     | 1.33625     | 359  | 791  | explicit |
| loss                     | 1.29195     | 360  | 793  | explicit |
| loss                     | 1.37848     | 361  | 795  | explicit |
| loss                     | 1.37728     | 362  | 797  | explicit |
| loss                     | 1.29842     | 363  | 799  | explicit |
| loss                     | 1.26567     | 364  | 801  | explicit |
| loss                     | 1.33102     | 365  | 803  | explicit |
| loss                     | 1.38719     | 366  | 805  | explicit |
| loss                     | 1.32539     | 367  | 807  | explicit |
| loss                     | 1.34273     | 368  | 809  | explicit |
| loss                     | 1.3113      | 369  | 811  | explicit |
| loss                     | 1.28381     | 370  | 813  | explicit |
| loss                     | 1.30328     | 371  | 815  | explicit |
| loss                     | 1.39278     | 372  | 817  | explicit |
| loss                     | 1.34293     | 373  | 819  | explicit |
| loss                     | 1.38617     | 374  | 821  | explicit |
| loss                     | 1.35478     | 375  | 823  | explicit |
| loss                     | 1.32229     | 376  | 825  | explicit |
| loss                     | 1.34435     | 377  | 827  | explicit |
| loss                     | 1.30054     | 378  | 829  | explicit |
| loss                     | 1.3888      | 379  | 831  | explicit |
| loss                     | 1.35706     | 380  | 833  | explicit |
| loss                     | 1.44664     | 381  | 835  | explicit |
| loss                     | 1.3056      | 382  | 837  | explicit |

| Metric                   | Value       | Step | Line | Source   |
|--------------------------|-------------|------|------|----------|
| loss                     | 1.24266     | 383  | 839  | explicit |
| loss                     | 1.26987     | 384  | 841  | explicit |
| loss                     | 1.32783     | 385  | 843  | explicit |
| loss                     | 1.35368     | 386  | 845  | explicit |
| loss                     | 1.33332     | 387  | 847  | explicit |
| loss                     | 1.3993      | 388  | 849  | explicit |
| loss                     | 1.32711     | 389  | 851  | explicit |
| loss                     | 1.33356     | 390  | 853  | explicit |
| loss                     | 1.3531      | 391  | 855  | explicit |
| loss                     | 1.30752     | 392  | 857  | explicit |
| loss                     | 1.32141     | 393  | 859  | explicit |
| loss                     | 1.3335      | 394  | 861  | explicit |
| loss                     | 1.33524     | 395  | 863  | explicit |
| loss                     | 1.37748     | 396  | 865  | explicit |
| loss                     | 1.25474     | 397  | 867  | explicit |
| loss                     | 1.275       | 398  | 869  | explicit |
| loss                     | 1.30899     | 399  | 871  | explicit |
| loss                     | 1.33885     | 400  | 873  | explicit |
| eval_loss                | 1.32483     | 400  | 876  | explicit |
| eval_runtime             | 22.1784     | 400  | 877  | explicit |
| eval_samples_per_second  | 16.232      | 400  | 878  | explicit |
| eval_steps_per_second    | 2.029       | 400  | 879  | explicit |
| eval_entropy             | 1.36021     | 400  | 880  | explicit |
| eval_num_tokens          | 1.22271e+07 | 400  | 881  | explicit |
| eval_mean_token_accuracy | 0.768343    | 400  | 882  | explicit |
| epoch                    | 1.97284     | 400  | 883  | explicit |
| loss                     | 1.36723     | 401  | 884  | explicit |
| loss                     | 1.2669      | 402  | 886  | explicit |
| loss                     | 1.28622     | 403  | 888  | explicit |
| loss                     | 1.3659      | 404  | 890  | explicit |
| loss                     | 1.30397     | 405  | 892  | explicit |
| loss                     | 1.38411     | 406  | 894  | explicit |
| loss                     | 1.29515     | 407  | 896  | explicit |
| loss                     | 1.20232     | 408  | 898  | explicit |
| loss                     | 1.32211     | 409  | 900  | explicit |
| loss                     | 1.24164     | 410  | 902  | explicit |
| loss                     | 1.31749     | 411  | 904  | explicit |
| loss                     | 1.29688     | 412  | 906  | explicit |
| loss                     | 1.32081     | 413  | 908  | explicit |
| loss                     | 1.33977     | 414  | 910  | explicit |
| loss                     | 1.34452     | 415  | 912  | explicit |
| loss                     | 1.32232     | 416  | 914  | explicit |
| loss                     | 1.3297      | 417  | 916  | explicit |
| loss                     | 1.33631     | 418  | 918  | explicit |
| loss                     | 1.32854     | 419  | 920  | explicit |
| loss                     | 1.36741     | 420  | 922  | explicit |
| loss                     | 1.28092     | 421  | 924  | explicit |
| loss                     | 1.3345      | 422  | 926  | explicit |
| loss                     | 1.3553      | 423  | 928  | explicit |
| loss                     | 1.34193     | 424  | 930  | explicit |

| Metric                   | Value       | Step | Line | Source   |
|--------------------------|-------------|------|------|----------|
| loss                     | 1.42417     | 425  | 932  | explicit |
| loss                     | 1.35213     | 426  | 934  | explicit |
| loss                     | 1.3853      | 427  | 936  | explicit |
| loss                     | 1.26153     | 428  | 938  | explicit |
| loss                     | 1.35889     | 429  | 940  | explicit |
| loss                     | 1.29597     | 430  | 942  | explicit |
| loss                     | 1.31341     | 431  | 944  | explicit |
| loss                     | 1.30257     | 432  | 946  | explicit |
| loss                     | 1.35322     | 433  | 948  | explicit |
| loss                     | 1.29995     | 434  | 950  | explicit |
| loss                     | 1.36613     | 435  | 952  | explicit |
| loss                     | 1.33392     | 436  | 954  | explicit |
| loss                     | 1.35453     | 437  | 956  | explicit |
| loss                     | 1.36887     | 438  | 958  | explicit |
| loss                     | 1.36776     | 439  | 960  | explicit |
| loss                     | 1.31035     | 440  | 962  | explicit |
| loss                     | 1.30112     | 441  | 964  | explicit |
| loss                     | 1.38712     | 442  | 966  | explicit |
| loss                     | 1.33896     | 443  | 968  | explicit |
| loss                     | 1.26031     | 444  | 970  | explicit |
| loss                     | 1.33834     | 445  | 972  | explicit |
| loss                     | 1.34874     | 446  | 974  | explicit |
| loss                     | 1.35472     | 447  | 976  | explicit |
| loss                     | 1.23554     | 448  | 978  | explicit |
| loss                     | 1.3445      | 449  | 980  | explicit |
| loss                     | 1.29363     | 450  | 982  | explicit |
| eval_loss                | 1.3255      | 450  | 985  | explicit |
| eval_runtime             | 22.1401     | 450  | 986  | explicit |
| eval_samples_per_second  | 16.26       | 450  | 987  | explicit |
| eval_steps_per_second    | 2.033       | 450  | 988  | explicit |
| eval_entropy             | 1.36203     | 450  | 989  | explicit |
| eval_num_tokens          | 1.37495e+07 | 450  | 990  | explicit |
| eval_mean_token_accuracy | 0.768086    | 450  | 991  | explicit |
| epoch                    | 2.21728     | 450  | 992  | explicit |
| loss                     | 1.3514      | 451  | 993  | explicit |
| loss                     | 1.31398     | 452  | 995  | explicit |
| loss                     | 1.34269     | 453  | 997  | explicit |
| loss                     | 1.26714     | 454  | 999  | explicit |
| loss                     | 1.29766     | 455  | 1001 | explicit |
| loss                     | 1.38899     | 456  | 1003 | explicit |
| loss                     | 1.27763     | 457  | 1005 | explicit |
| loss                     | 1.33767     | 458  | 1007 | explicit |
| loss                     | 1.36222     | 459  | 1009 | explicit |
| loss                     | 1.29712     | 460  | 1011 | explicit |
| loss                     | 1.31692     | 461  | 1013 | explicit |
| loss                     | 1.31385     | 462  | 1015 | explicit |
| loss                     | 1.32936     | 463  | 1017 | explicit |
| loss                     | 1.30666     | 464  | 1019 | explicit |
| loss                     | 1.35176     | 465  | 1021 | explicit |
| loss                     | 1.26962     | 466  | 1023 | explicit |

| Metric                   | Value       | Step     | Line     | Source   |     |
|--------------------------|-------------|----------|----------|----------|-----|
| loss                     | 1.3144      | 467      | 1025     | explicit |     |
| loss                     | 1.30339     | 468      | 1027     | explicit |     |
| loss                     | 1.33172     | 469      | 1029     | explicit |     |
| loss                     | 1.36505     | 470      | 1031     | explicit |     |
| loss                     | 1.26197     | 471      | 1033     | explicit |     |
| loss                     | 1.27726     | 472      | 1035     | explicit |     |
| loss                     | 1.32968     | 473      | 1037     | explicit |     |
| loss                     | 1.28692     | 474      | 1039     | explicit |     |
| loss                     | 1.31702     | 475      | 1041     | explicit |     |
| loss                     | 1.32109     | 476      | 1043     | explicit |     |
| loss                     | 1.32393     | 477      | 1045     | explicit |     |
| loss                     | 1.37259     | 478      | 1047     | explicit |     |
| loss                     | 1.3351      | 479      | 1049     | explicit |     |
| loss                     | 1.40239     | 480      | 1051     | explicit |     |
| loss                     | 1.34091     | 481      | 1053     | explicit |     |
| loss                     | 1.26832     | 482      | 1055     | explicit |     |
| loss                     | 1.40606     | 483      | 1057     | explicit |     |
| loss                     | 1.35638     | 484      | 1059     | explicit |     |
| loss                     | 1.28052     | 485      | 1061     | explicit |     |
| loss                     | 1.29174     | 486      | 1063     | explicit |     |
| loss                     | 1.39009     | 487      | 1065     | explicit |     |
| loss                     | 1.35829     | 488      | 1067     | explicit |     |
| loss                     | 1.29894     | 489      | 1069     | explicit |     |
| loss                     | 1.2273      | 490      | 1071     | explicit |     |
| loss                     | 1.37781     | 491      | 1073     | explicit |     |
| loss                     | 1.29309     | 492      | 1075     | explicit |     |
| loss                     | 1.38396     | 493      | 1077     | explicit |     |
| loss                     | 1.34988     | 494      | 1079     | explicit |     |
| loss                     | 1.36201     | 495      | 1081     | explicit |     |
| loss                     | 1.36121     | 496      | 1083     | explicit |     |
| loss                     | 1.40236     | 497      | 1085     | explicit |     |
| loss                     | 1.37914     | 498      | 1087     | explicit |     |
| loss                     | 1.35373     | 499      | 1089     | explicit |     |
| loss                     | 1.30145     | 500      | 1091     | explicit |     |
| eval_loss                | 1.32598     | 500      | 1094     | explicit |     |
| eval_runtime             | 22.1618     | 500      | 1095     | explicit |     |
| eval_samples_per_second  | 16.244      | 500      | 1096     | explicit |     |
| eval_steps_per_second    | 2.031       | 500      | 1097     | explicit |     |
| eval_entropy             | 1.36176     | 500      | 1098     | explicit |     |
| eval_num_tokens          | 1.52793e+07 | 500      | 1099     | explicit |     |
| eval_mean_token_accuracy | 0.768744    | 500      | 1100     | explicit |     |
| epoch                    | 2.4642      | 500      | 1101     | explicit |     |
|                          |             |          |          |          |     |
| Phase                    | Line        | Time     | CPU time | Memory   | GPU |
| train                    | 3           | 05:46:09 | 2.3 s    | 0.79 GB  |     |

**Run 4: sft.yaml**

| <b>Metric</b> | <b>Value</b> | <b>Step</b> | <b>Line</b> | <b>Source</b> |
|---------------|--------------|-------------|-------------|---------------|
| phi_bam       | 0.823658     | 0           | 11          | explicit      |
| mer_bam       | 0            | 0           | 12          | explicit      |
| mcs_bam       | 0            | 0           | 13          | explicit      |
| uwec_bam      | 1            | 0           | 14          | explicit      |
| phi_bmq       | 0.752429     | 0           | 15          | explicit      |
| mer_bmq       | 3            | 0           | 16          | explicit      |
| mcs_bmq       | 0.833333     | 0           | 17          | explicit      |
| uwec_bmq      | 0.666667     | 0           | 18          | explicit      |
| phi_boz       | 0.916823     | 0           | 19          | explicit      |
| mer_boz       | 0            | 0           | 20          | explicit      |
| mcs_boz       | 0            | 0           | 21          | explicit      |
| uwec_boz      | 1            | 0           | 22          | explicit      |
| phi_dtm       | 0.809387     | 0           | 23          | explicit      |
| mer_dtm       | 0.5          | 0           | 24          | explicit      |
| mcs_dtm       | 0.333333     | 0           | 25          | explicit      |
| uwec_dtm      | 0.666667     | 0           | 26          | explicit      |
| phi_ful       | 0.980054     | 0           | 27          | explicit      |
| mer_ful       | 3            | 0           | 28          | explicit      |
| mcs_ful       | 0.75         | 0           | 29          | explicit      |
| uwec_ful      | 0.625        | 0           | 30          | explicit      |
| phi_mey       | 0.768726     | 0           | 31          | explicit      |
| mer_mey       | 0            | 0           | 32          | explicit      |
| mcs_mey       | 0            | 0           | 33          | explicit      |
| uwec_mey      | 0.5          | 0           | 34          | explicit      |
| phi_mku       | 0.908601     | 0           | 35          | explicit      |
| mer_mku       | 12           | 0           | 36          | explicit      |
| mcs_mku       | 0.923077     | 0           | 37          | explicit      |
| uwec_mku      | 0.538462     | 0           | 38          | explicit      |
| phi_kao       | 0.970238     | 0           | 39          | explicit      |
| mer_kao       | 25.3333      | 0           | 40          | explicit      |
| mcs_kao       | 0.974359     | 0           | 41          | explicit      |
| uwec_kao      | 1            | 0           | 42          | explicit      |
| phi_myk       | 0.7953       | 0           | 43          | explicit      |
| mer_myk       | 10           | 0           | 44          | explicit      |
| mcs_myk       | 0.909091     | 0           | 45          | explicit      |
| uwec_myk      | 0.5          | 0           | 46          | explicit      |
| phi_ses       | 0.836952     | 0           | 47          | explicit      |
| mer_ses       | 0            | 0           | 48          | explicit      |
| mcs_ses       | 0            | 0           | 49          | explicit      |
| uwec_ses      | 1            | 0           | 50          | explicit      |
| phi_spp       | 0.853018     | 0           | 51          | explicit      |
| mer_spp       | 6.66667      | 0           | 52          | explicit      |
| mcs_spp       | 0.909091     | 0           | 53          | explicit      |
| uwec_spp      | 0.545455     | 0           | 54          | explicit      |
| phi_snk       | 0.781673     | 0           | 55          | explicit      |
| mer_snk       | 0.333333     | 0           | 56          | explicit      |
| mcs_snk       | 0.333333     | 0           | 57          | explicit      |
| uwec_snk      | 0.666667     | 0           | 58          | explicit      |

| Metric   | Value    | Step | Line | Source   |
|----------|----------|------|------|----------|
| phi_taq  | 0.84844  | 0    | 59   | explicit |
| mer_taq  | 0.5      | 0    | 60   | explicit |
| mcs_taq  | 0.5      | 0    | 61   | explicit |
| uwec_taq | 1        | 0    | 62   | explicit |
| phi      | 0.849638 | 0    | 63   | explicit |
| mer      | 5.16667  | 0    | 65   | explicit |
| mcs      | 0.865169 | 0    | 67   | explicit |
| uwec     | 0.769663 | 0    | 69   | explicit |

  

| Phase | Line | Time     | CPU time | Memory  | GPU |
|-------|------|----------|----------|---------|-----|
| eval  | 2    | 06:19:32 | 2.3 s    | 0.73 GB |     |

## B Provenance timeline and activity map

### Run 1: kveritas\_arm.py

| # | Snapshot  | Time     | Root             | Changed |
|---|-----------|----------|------------------|---------|
| 0 | run_start | 05:45:09 | 0a1f37f88004b036 | 134     |
| 1 | run_end   | 05:45:11 | 32cfabf37bf320ae | 1       |

  

| Event | Path or command                                     | Hash |
|-------|-----------------------------------------------------|------|
| read  | cookbooks/kveritas_arm.py                           |      |
| read  | beni/utils/res/mdf_fields.json                      |      |
| read  | beni/utils/res/language_metadata.json               |      |
| read  | beni/utils/res/iso_code_639_3.json                  |      |
| read  | beni/cli/main.py                                    |      |
| read  | beni/core/__init__.py                               |      |
| read  | beni/core/language.py                               |      |
| read  | beni/core/srl/config.py                             |      |
| read  | beni/core/srl/unified.py                            |      |
| read  | beni/core/safety/__init__.py                        |      |
| read  | .github/workflows/docs.yml                          |      |
| read  | __init__.py                                         |      |
| read  | beni/__init__.py                                    |      |
| read  | beni/cli/__init__.py                                |      |
| read  | beni/core/compute/__init__.py                       |      |
| read  | beni/core/compute/helpers.py                        |      |
| read  | beni/core/compute/metrics.py                        |      |
| read  | beni/core/compute/rewards.py                        |      |
| read  | beni/core/compute/uwec.py                           |      |
| read  | beni/core/hub/__init__.py                           |      |
| read  | beni/core/hub/model_card.py                         |      |
| read  | beni/core/morphotactic/__init__.py                  |      |
| read  | beni/core/morphotactic/dabax.py                     |      |
| read  | beni/core/morphotactic/distil/__init__.py           |      |
| read  | beni/core/morphotactic/distil/distillation.py       |      |
| read  | beni/core/morphotactic/distil/providers/__init__.py |      |
| read  | beni/core/morphotactic/distil/providers/base.py     |      |
| read  | beni/core/morphotactic/distil/providers/gguf.py     |      |
| read  | beni/core/morphotactic/distil/providers/google.py   |      |

| Event | Path or command                                          | Hash |
|-------|----------------------------------------------------------|------|
| read  | beni/core/morphotactic/distil/providers/openai_compat.py |      |
| read  | beni/core/pipeline.py                                    |      |
| read  | beni/core/safety/governor.py                             |      |
| read  | beni/core/safety/spec.py                                 |      |
| read  | beni/core/srl/__init__.py                                |      |
| read  | beni/core/srl/algorithm1.py                              |      |
| read  | beni/core/srl/apo/__init__.py                            |      |
| read  | beni/core/srl/apo/apo.py                                 |      |
| read  | beni/core/srl/dpo/__init__.py                            |      |
| read  | beni/core/srl/dpo/dpo.py                                 |      |
| read  | beni/core/srl/grpo/__init__.py                           |      |
| read  | beni/core/srl/grpo/callbacks.py                          |      |
| read  | beni/core/srl/grpo/grpo.py                               |      |
| read  | beni/core/srl/jax/__init__.py                            |      |
| read  | beni/core/srl/jax/policy.py                              |      |
| read  | beni/core/srl/plugin.py                                  |      |
| read  | beni/core/srl/sft/__init__.py                            |      |
| read  | beni/core/srl/sft/sft.py                                 |      |
| read  | beni/core/wordfreq/__init__.py                           |      |
| read  | beni/data/__init__.py                                    |      |
| read  | beni/data/baselines/bmq/ex.json                          |      |
| read  | beni/data/datasets.py                                    |      |
| read  | beni/data/exp.yaml                                       |      |
| read  | beni/data/test.json                                      |      |
| read  | beni/utis/__init__.py                                    |      |
| read  | beni/utis/config.py                                      |      |
| read  | beni/utis/hf_lang.py                                     |      |
| read  | beni/utis/res/__init__.py                                |      |
| read  | cleanup.sh                                               |      |
| read  | configs/apo_bam.yaml                                     |      |
| read  | configs/dpo_bam.yaml                                     |      |
| read  | configs/exp.yaml                                         |      |
| read  | configs/grpo_bam.yaml                                    |      |
| read  | configs/grpo_multilang.yaml                              |      |
| read  | configs/matrix/base_multi13.yaml                         |      |
| read  | configs/matrix/models.json                               |      |
| read  | configs/presets/multi13.yaml                             |      |
| read  | configs/presets/single.yaml                              |      |
| read  | conftest.py                                              |      |
| read  | cookbooks/eval_apo.ipynb                                 |      |
| read  | cookbooks/eval_dpo.ipynb                                 |      |
| read  | cookbooks/eval_grpo.ipynb                                |      |
| read  | cookbooks/eval_sft.ipynb                                 |      |
| read  | cookbooks/experimentation_matrix.ipynb                   |      |
| read  | cookbooks/run_exp.ipynb                                  |      |
| read  | docs/javascripts/mathjax.js                              |      |
| read  | docs/javascripts/typewriter.js                           |      |
| read  | experiments/matrix/gemma-3-270m_apo/config.yaml          |      |
| read  | experiments/matrix/gemma-3-270m_dpo/config.yaml          |      |
| read  | experiments/matrix/gemma-3-270m_grpo/config.yaml         |      |

| Event | Path or command                                           | Hash |
|-------|-----------------------------------------------------------|------|
| read  | experiments/matrix/gemma-3-270m_sft/config.yaml           |      |
| read  | experiments/matrix/gemma-4-e2b_apo/config.yaml            |      |
| read  | experiments/matrix/gemma-4-e2b_dpo/config.yaml            |      |
| read  | experiments/matrix/gemma-4-e2b_grpo/config.yaml           |      |
| read  | experiments/matrix/gemma-4-e2b_sft/config.yaml            |      |
| read  | experiments/matrix/gemma-4-e4b_apo/config.yaml            |      |
| read  | experiments/matrix/gemma-4-e4b_dpo/config.yaml            |      |
| read  | experiments/matrix/gemma-4-e4b_grpo/config.yaml           |      |
| read  | experiments/matrix/gemma-4-e4b_sft/config.yaml            |      |
| read  | experiments/matrix/granite-4.0-h-350m_apo/config.yaml     |      |
| read  | experiments/matrix/granite-4.0-h-350m_dpo/config.yaml     |      |
| read  | experiments/matrix/granite-4.0-h-350m_grpo/config.yaml    |      |
| read  | experiments/matrix/granite-4.0-h-350m_sft/config.yaml     |      |
| read  | experiments/matrix/lfm2.5-230m_apo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-230m_dpo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-230m_grpo/config.yaml           |      |
| read  | experiments/matrix/lfm2.5-230m_sft/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-350m_apo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-350m_dpo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-350m_grpo/config.yaml           |      |
| read  | experiments/matrix/lfm2.5-350m_sft/config.yaml            |      |
| read  | experiments/matrix/matrix_results.json                    |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_apo/config.yaml  |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_dpo/config.yaml  |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_grpo/config.yaml |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_sft/config.yaml  |      |
| read  | experiments/matrix/qwen2.5-0.5b_apo/config.yaml           |      |
| read  | experiments/matrix/qwen2.5-0.5b_dpo/config.yaml           |      |
| read  | experiments/matrix/qwen2.5-0.5b_grpo/config.yaml          |      |
| read  | experiments/matrix/qwen2.5-0.5b_sft/config.yaml           |      |
| read  | experiments/matrix/qwen3-0.6b_apo/config.yaml             |      |
| read  | experiments/matrix/qwen3-0.6b_dpo/config.yaml             |      |
| read  | experiments/matrix/qwen3-0.6b_grpo/config.yaml            |      |
| read  | experiments/matrix/qwen3-0.6b_sft/config.yaml             |      |
| read  | experiments/matrix/smollm2-135m_apo/config.yaml           |      |
| read  | experiments/matrix/smollm2-135m_dpo/config.yaml           |      |
| read  | experiments/matrix/smollm2-135m_grpo/config.yaml          |      |
| read  | experiments/matrix/smollm2-135m_sft/config.yaml           |      |
| read  | experiments/matrix/smollm3-3b_apo/config.yaml             |      |
| read  | experiments/matrix/smollm3-3b_dpo/config.yaml             |      |
| read  | experiments/matrix/smollm3-3b_grpo/config.yaml            |      |
| read  | experiments/matrix/smollm3-3b_sft/config.yaml             |      |
| read  | experiments/matrix/tinyllama-1.1b_apo/config.yaml         |      |
| read  | experiments/matrix/tinyllama-1.1b_dpo/config.yaml         |      |
| read  | experiments/matrix/tinyllama-1.1b_grpo/config.yaml        |      |
| read  | experiments/matrix/tinyllama-1.1b_sft/config.yaml         |      |
| read  | mkdocs.yml                                                |      |
| read  | pyproject.toml                                            |      |
| read  | scripts/run_matrix.py                                     |      |
| read  | setup.cfg                                                 |      |

| Event   | Path or command                                                                 | Hash             |
|---------|---------------------------------------------------------------------------------|------------------|
| read    | setup.py                                                                        |                  |
| read    | tests/__init__.py                                                               |                  |
| read    | tests/test_framework.py                                                         |                  |
| read    | tests/test_grpo_pipeline.py                                                     |                  |
| read    | tests/test_kveritas_callback.py                                                 |                  |
| write   | cookbooks/kveritas/sft.yaml                                                     | 0ae0e696903bee37 |
| process | /tmp/kv-sebeni/venv/bin/python3.12 /tmp/kv-sebeni/venv/bin/sebeni init --lan... |                  |

**Run 2: sft.yaml**

| # | Snapshot      | Time     | Root             | Changed |
|---|---------------|----------|------------------|---------|
| 0 | run_start     | 05:45:12 | 32cfabf37bf320ae | 135     |
| 1 | phase distill | 05:45:13 | 32cfabf37bf320ae | 0       |
| 2 | run_end       | 05:46:06 | 32cfabf37bf320ae | 0       |

| Event | Path or command                                          | Hash |
|-------|----------------------------------------------------------|------|
| read  | beni/__init__.py                                         |      |
| read  | beni/utis/__init__.py                                    |      |
| read  | beni/utis/config.py                                      |      |
| read  | beni/utis/res/mdf_fields.json                            |      |
| read  | beni/utis/res/language_metadata.json                     |      |
| read  | beni/utis/res/iso_code_639_3.json                        |      |
| read  | beni/core/srl/config.py                                  |      |
| read  | beni/core/srl/unified.py                                 |      |
| read  | beni/core/safety/__init__.py                             |      |
| read  | beni/core/safety/governor.py                             |      |
| read  | .github/workflows/docs.yml                               |      |
| read  | __init__.py                                              |      |
| read  | beni/cli/__init__.py                                     |      |
| read  | beni/cli/main.py                                         |      |
| read  | beni/core/__init__.py                                    |      |
| read  | beni/core/compute/helpers.py                             |      |
| read  | beni/core/compute/metrics.py                             |      |
| read  | beni/core/compute/rewards.py                             |      |
| read  | beni/core/compute/uwec.py                                |      |
| read  | beni/core/hub/__init__.py                                |      |
| read  | beni/core/hub/model_card.py                              |      |
| read  | beni/core/language.py                                    |      |
| read  | beni/core/morphotactic/dabax.py                          |      |
| read  | beni/core/morphotactic/distil/__init__.py                |      |
| read  | beni/core/morphotactic/distil/providers/__init__.py      |      |
| read  | beni/core/morphotactic/distil/providers/gguf.py          |      |
| read  | beni/core/morphotactic/distil/providers/google.py        |      |
| read  | beni/core/morphotactic/distil/providers/openai_compat.py |      |
| read  | beni/core/safety/spec.py                                 |      |
| read  | beni/core/srl/__init__.py                                |      |
| read  | beni/core/srl/algorithm1.py                              |      |
| read  | beni/core/srl/apo/__init__.py                            |      |
| read  | beni/core/srl/apo/apo.py                                 |      |
| read  | beni/core/srl/dpo/__init__.py                            |      |
| read  | beni/core/srl/dpo/dpo.py                                 |      |

| Event | Path or command                                       | Hash |
|-------|-------------------------------------------------------|------|
| read  | beni/core/srl/grpo/__init__.py                        |      |
| read  | beni/core/srl/grpo/callbacks.py                       |      |
| read  | beni/core/srl/grpo/grpo.py                            |      |
| read  | beni/core/srl/jax/__init__.py                         |      |
| read  | beni/core/srl/jax/policy.py                           |      |
| read  | beni/core/srl/plugin.py                               |      |
| read  | beni/core/srl/sft/__init__.py                         |      |
| read  | beni/core/srl/sft/sft.py                              |      |
| read  | beni/core/wordfreq/__init__.py                        |      |
| read  | beni/data/__init__.py                                 |      |
| read  | beni/data/baselines/bmq/ex.json                       |      |
| read  | beni/data/datasets.py                                 |      |
| read  | beni/data/exp.yaml                                    |      |
| read  | beni/data/test.json                                   |      |
| read  | beni/utlis/hf_lang.py                                 |      |
| read  | beni/utlis/res/__init__.py                            |      |
| read  | cleanup.sh                                            |      |
| read  | configs/apo_bam.yaml                                  |      |
| read  | configs/dpo_bam.yaml                                  |      |
| read  | configs/exp.yaml                                      |      |
| read  | configs/grpo_bam.yaml                                 |      |
| read  | configs/grpo_multilang.yaml                           |      |
| read  | configs/matrix/base_multi13.yaml                      |      |
| read  | configs/matrix/models.json                            |      |
| read  | configs/presets/multi13.yaml                          |      |
| read  | configs/presets/single.yaml                           |      |
| read  | conftest.py                                           |      |
| read  | cookbooks/eval_apo.ipynb                              |      |
| read  | cookbooks/eval_dpo.ipynb                              |      |
| read  | cookbooks/eval_grpo.ipynb                             |      |
| read  | cookbooks/eval_sft.ipynb                              |      |
| read  | cookbooks/experimentation_matrix.ipynb                |      |
| read  | cookbooks/kveritas/sft.yaml                           |      |
| read  | cookbooks/kveritas_arm.py                             |      |
| read  | cookbooks/run_exp.ipynb                               |      |
| read  | docs/javascripts/mathjax.js                           |      |
| read  | docs/javascripts/typewriter.js                        |      |
| read  | experiments/matrix/gemma-3-270m_apo/config.yaml       |      |
| read  | experiments/matrix/gemma-3-270m_dpo/config.yaml       |      |
| read  | experiments/matrix/gemma-3-270m_grpo/config.yaml      |      |
| read  | experiments/matrix/gemma-3-270m_sft/config.yaml       |      |
| read  | experiments/matrix/gemma-4-e2b_apo/config.yaml        |      |
| read  | experiments/matrix/gemma-4-e2b_dpo/config.yaml        |      |
| read  | experiments/matrix/gemma-4-e2b_grpo/config.yaml       |      |
| read  | experiments/matrix/gemma-4-e2b_sft/config.yaml        |      |
| read  | experiments/matrix/gemma-4-e4b_apo/config.yaml        |      |
| read  | experiments/matrix/gemma-4-e4b_dpo/config.yaml        |      |
| read  | experiments/matrix/gemma-4-e4b_grpo/config.yaml       |      |
| read  | experiments/matrix/gemma-4-e4b_sft/config.yaml        |      |
| read  | experiments/matrix/granite-4.0-h-350m_apo/config.yaml |      |

| Event | Path or command                                           | Hash |
|-------|-----------------------------------------------------------|------|
| read  | experiments/matrix/granite-4.0-h-350m_dpo/config.yaml     |      |
| read  | experiments/matrix/granite-4.0-h-350m_grpo/config.yaml    |      |
| read  | experiments/matrix/granite-4.0-h-350m_sft/config.yaml     |      |
| read  | experiments/matrix/lfm2.5-230m_apo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-230m_dpo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-230m_grpo/config.yaml           |      |
| read  | experiments/matrix/lfm2.5-230m_sft/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-350m_apo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-350m_dpo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-350m_grpo/config.yaml           |      |
| read  | experiments/matrix/lfm2.5-350m_sft/config.yaml            |      |
| read  | experiments/matrix/matrix_results.json                    |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_apo/config.yaml  |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_dpo/config.yaml  |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_grpo/config.yaml |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_sft/config.yaml  |      |
| read  | experiments/matrix/qwen2.5-0.5b_apo/config.yaml           |      |
| read  | experiments/matrix/qwen2.5-0.5b_dpo/config.yaml           |      |
| read  | experiments/matrix/qwen2.5-0.5b_grpo/config.yaml          |      |
| read  | experiments/matrix/qwen2.5-0.5b_sft/config.yaml           |      |
| read  | experiments/matrix/qwen3-0.6b_apo/config.yaml             |      |
| read  | experiments/matrix/qwen3-0.6b_dpo/config.yaml             |      |
| read  | experiments/matrix/qwen3-0.6b_grpo/config.yaml            |      |
| read  | experiments/matrix/qwen3-0.6b_sft/config.yaml             |      |
| read  | experiments/matrix/smollm2-135m_apo/config.yaml           |      |
| read  | experiments/matrix/smollm2-135m_dpo/config.yaml           |      |
| read  | experiments/matrix/smollm2-135m_grpo/config.yaml          |      |
| read  | experiments/matrix/smollm2-135m_sft/config.yaml           |      |
| read  | experiments/matrix/smollm3-3b_apo/config.yaml             |      |
| read  | experiments/matrix/smollm3-3b_dpo/config.yaml             |      |
| read  | experiments/matrix/smollm3-3b_grpo/config.yaml            |      |
| read  | experiments/matrix/smollm3-3b_sft/config.yaml             |      |
| read  | experiments/matrix/tinyllama-1.1b_apo/config.yaml         |      |
| read  | experiments/matrix/tinyllama-1.1b_dpo/config.yaml         |      |
| read  | experiments/matrix/tinyllama-1.1b_grpo/config.yaml        |      |
| read  | experiments/matrix/tinyllama-1.1b_sft/config.yaml         |      |
| read  | mkdocs.yml                                                |      |
| read  | pyproject.toml                                            |      |
| read  | scripts/run_matrix.py                                     |      |
| read  | setup.cfg                                                 |      |
| read  | setup.py                                                  |      |
| read  | tests/__init__.py                                         |      |
| read  | tests/test_framework.py                                   |      |
| read  | tests/test_grpo_pipeline.py                               |      |
| read  | tests/test_kveritas_callback.py                           |      |
| read  | beni/core/pipeline.py                                     |      |
| read  | beni/data/raw/dataset_300_samples.jsonl                   |      |
| read  | beni/core/morphotactic/__init__.py                        |      |
| read  | beni/core/morphotactic/distil/distillation.py             |      |
| read  | beni/core/morphotactic/distil/providers/base.py           |      |

| Event | Path or command                       | Hash |
|-------|---------------------------------------|------|
| read  | beni/core/compute/__init__.py         |      |
| read  | beni/data/baselines/spp/baseline.gram |      |
| read  | beni/data/baselines/spp/baseline.dict |      |
| read  | beni/data/baselines/bam/baseline.gram |      |
| read  | beni/data/baselines/bam/baseline.dict |      |
| read  | beni/data/baselines/taq/baseline.gram |      |
| read  | beni/data/baselines/taq/baseline.dict |      |
| read  | beni/data/baselines/bmq/baseline.gram |      |
| read  | beni/data/baselines/bmq/baseline.dict |      |
| read  | beni/data/baselines/mku/baseline.gram |      |
| read  | beni/data/baselines/mku/baseline.dict |      |
| read  | beni/data/baselines/dtm/baseline.gram |      |
| read  | beni/data/baselines/dtm/baseline.dict |      |
| read  | beni/data/baselines/myk/baseline.gram |      |
| read  | beni/data/baselines/myk/baseline.dict |      |
| read  | beni/data/baselines/boz/baseline.gram |      |
| read  | beni/data/baselines/boz/baseline.dict |      |
| read  | beni/data/baselines/ses/baseline.gram |      |
| read  | beni/data/baselines/ses/baseline.dict |      |
| read  | beni/data/baselines/ful/baseline.gram |      |
| read  | beni/data/baselines/ful/baseline.dict |      |
| read  | beni/data/baselines/snk/baseline.gram |      |
| read  | beni/data/baselines/snk/baseline.dict |      |

**Run 3: 0.1**

| # | Snapshot    | Time     | Root             | Changed |
|---|-------------|----------|------------------|---------|
| 0 | run_start   | 05:46:07 | 32cfabf37bf320ae | 135     |
| 1 | phase train | 05:46:09 | 32cfabf37bf320ae | 0       |
| 2 | run_end     | 06:19:30 | 32cfabf37bf320ae | 0       |

| Event | Path or command                       | Hash |
|-------|---------------------------------------|------|
| read  | beni/core/__init__.py                 |      |
| read  | beni/core/language.py                 |      |
| read  | beni/utils/config.py                  |      |
| read  | beni/utils/res/mdf_fields.json        |      |
| read  | beni/utils/res/language_metadata.json |      |
| read  | beni/utils/res/iso_code_639_3.json    |      |
| read  | beni/core/srl/__init__.py             |      |
| read  | beni/core/srl/config.py               |      |
| read  | beni/core/safety/__init__.py          |      |
| read  | .github/workflows/docs.yml            |      |
| read  | __init__.py                           |      |
| read  | beni/__init__.py                      |      |
| read  | beni/cli/__init__.py                  |      |
| read  | beni/cli/main.py                      |      |
| read  | beni/core/compute/__init__.py         |      |
| read  | beni/core/compute/helpers.py          |      |
| read  | beni/core/compute/metrics.py          |      |
| read  | beni/core/compute/rewards.py          |      |
| read  | beni/core/compute/uwec.py             |      |

| Event | Path or command                                          | Hash |
|-------|----------------------------------------------------------|------|
| read  | beni/core/hub/__init__.py                                |      |
| read  | beni/core/hub/model_card.py                              |      |
| read  | beni/core/morphotactic/__init__.py                       |      |
| read  | beni/core/morphotactic/dabax.py                          |      |
| read  | beni/core/morphotactic/distil/__init__.py                |      |
| read  | beni/core/morphotactic/distil/distillation.py            |      |
| read  | beni/core/morphotactic/distil/providers/gguf.py          |      |
| read  | beni/core/morphotactic/distil/providers/google.py        |      |
| read  | beni/core/morphotactic/distil/providers/openai_compat.py |      |
| read  | beni/core/safety/governor.py                             |      |
| read  | beni/core/srl/algorithm1.py                              |      |
| read  | beni/core/srl/grpo/__init__.py                           |      |
| read  | beni/core/srl/grpo/callbacks.py                          |      |
| read  | beni/core/srl/jax/__init__.py                            |      |
| read  | beni/core/srl/jax/policy.py                              |      |
| read  | beni/core/srl/sft/sft.py                                 |      |
| read  | beni/core/srl/unified.py                                 |      |
| read  | beni/core/wordfreq/__init__.py                           |      |
| read  | beni/data/__init__.py                                    |      |
| read  | beni/data/baselines/bmq/ex.json                          |      |
| read  | beni/data/datasets.py                                    |      |
| read  | beni/data/exp.yaml                                       |      |
| read  | beni/data/test.json                                      |      |
| read  | beni/utis/__init__.py                                    |      |
| read  | beni/utis/res/__init__.py                                |      |
| read  | cleanup.sh                                               |      |
| read  | configs/apo_bam.yaml                                     |      |
| read  | configs/dpo_bam.yaml                                     |      |
| read  | configs/exp.yaml                                         |      |
| read  | configs/grpo_bam.yaml                                    |      |
| read  | configs/grpo_multilang.yaml                              |      |
| read  | configs/matrix/base_multi13.yaml                         |      |
| read  | configs/matrix/models.json                               |      |
| read  | configs/presets/multi13.yaml                             |      |
| read  | configs/presets/single.yaml                              |      |
| read  | conftest.py                                              |      |
| read  | cookbooks/eval_apo.ipynb                                 |      |
| read  | cookbooks/eval_dpo.ipynb                                 |      |
| read  | cookbooks/eval_grpo.ipynb                                |      |
| read  | cookbooks/eval_sft.ipynb                                 |      |
| read  | cookbooks/experimentation_matrix.ipynb                   |      |
| read  | cookbooks/kveritas/sft.yaml                              |      |
| read  | cookbooks/kveritas_arm.py                                |      |
| read  | cookbooks/run_exp.ipynb                                  |      |
| read  | docs/javascripts/mathjax.js                              |      |
| read  | docs/javascripts/typewriter.js                           |      |
| read  | experiments/matrix/gemma-3-270m_apo/config.yaml          |      |
| read  | experiments/matrix/gemma-3-270m_dpo/config.yaml          |      |
| read  | experiments/matrix/gemma-3-270m_grpo/config.yaml         |      |
| read  | experiments/matrix/gemma-3-270m_sft/config.yaml          |      |

| Event | Path or command                                           | Hash |
|-------|-----------------------------------------------------------|------|
| read  | experiments/matrix/gemma-4-e2b_apo/config.yaml            |      |
| read  | experiments/matrix/gemma-4-e2b_dpo/config.yaml            |      |
| read  | experiments/matrix/gemma-4-e2b_grpo/config.yaml           |      |
| read  | experiments/matrix/gemma-4-e2b_sft/config.yaml            |      |
| read  | experiments/matrix/gemma-4-e4b_apo/config.yaml            |      |
| read  | experiments/matrix/gemma-4-e4b_dpo/config.yaml            |      |
| read  | experiments/matrix/gemma-4-e4b_grpo/config.yaml           |      |
| read  | experiments/matrix/gemma-4-e4b_sft/config.yaml            |      |
| read  | experiments/matrix/granite-4.0-h-350m_apo/config.yaml     |      |
| read  | experiments/matrix/granite-4.0-h-350m_dpo/config.yaml     |      |
| read  | experiments/matrix/granite-4.0-h-350m_grpo/config.yaml    |      |
| read  | experiments/matrix/granite-4.0-h-350m_sft/config.yaml     |      |
| read  | experiments/matrix/lfm2.5-230m_apo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-230m_dpo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-230m_grpo/config.yaml           |      |
| read  | experiments/matrix/lfm2.5-230m_sft/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-350m_apo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-350m_dpo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-350m_grpo/config.yaml           |      |
| read  | experiments/matrix/lfm2.5-350m_sft/config.yaml            |      |
| read  | experiments/matrix/matrix_results.json                    |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_apo/config.yaml  |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_dpo/config.yaml  |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_grpo/config.yaml |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_sft/config.yaml  |      |
| read  | experiments/matrix/qwen2.5-0.5b_apo/config.yaml           |      |
| read  | experiments/matrix/qwen2.5-0.5b_dpo/config.yaml           |      |
| read  | experiments/matrix/qwen2.5-0.5b_grpo/config.yaml          |      |
| read  | experiments/matrix/qwen2.5-0.5b_sft/config.yaml           |      |
| read  | experiments/matrix/qwen3-0.6b_apo/config.yaml             |      |
| read  | experiments/matrix/qwen3-0.6b_dpo/config.yaml             |      |
| read  | experiments/matrix/qwen3-0.6b_grpo/config.yaml            |      |
| read  | experiments/matrix/qwen3-0.6b_sft/config.yaml             |      |
| read  | experiments/matrix/smollm2-135m_apo/config.yaml           |      |
| read  | experiments/matrix/smollm2-135m_dpo/config.yaml           |      |
| read  | experiments/matrix/smollm2-135m_grpo/config.yaml          |      |
| read  | experiments/matrix/smollm2-135m_sft/config.yaml           |      |
| read  | experiments/matrix/smollm3-3b_apo/config.yaml             |      |
| read  | experiments/matrix/smollm3-3b_dpo/config.yaml             |      |
| read  | experiments/matrix/smollm3-3b_grpo/config.yaml            |      |
| read  | experiments/matrix/smollm3-3b_sft/config.yaml             |      |
| read  | experiments/matrix/tinyllama-1.1b_apo/config.yaml         |      |
| read  | experiments/matrix/tinyllama-1.1b_dpo/config.yaml         |      |
| read  | experiments/matrix/tinyllama-1.1b_grpo/config.yaml        |      |
| read  | experiments/matrix/tinyllama-1.1b_sft/config.yaml         |      |
| read  | mkdocs.yml                                                |      |
| read  | pyproject.toml                                            |      |
| read  | scripts/run_matrix.py                                     |      |
| read  | setup.cfg                                                 |      |
| read  | setup.py                                                  |      |

| Event   | Path or command                                     | Hash |
|---------|-----------------------------------------------------|------|
| read    | tests/__init__.py                                   |      |
| read    | tests/test_framework.py                             |      |
| read    | tests/test_grpo_pipeline.py                         |      |
| read    | tests/test_kveritas_callback.py                     |      |
| read    | beni/core/safety/spec.py                            |      |
| read    | beni/data/raw/dataset_300_samples.jsonl             |      |
| read    | beni/core/pipeline.py                               |      |
| read    | beni/core/srl/grpo/grpo.py                          |      |
| read    | beni/core/srl/plugin.py                             |      |
| read    | beni/utils/hf_lang.py                               |      |
| read    | beni/core/morphotactic/distil/providers/__init__.py |      |
| read    | beni/core/morphotactic/distil/providers/base.py     |      |
| read    | beni/core/srl/dpo/__init__.py                       |      |
| read    | beni/core/srl/dpo/dpo.py                            |      |
| read    | beni/core/srl/apo/__init__.py                       |      |
| read    | beni/core/srl/apo/apo.py                            |      |
| read    | beni/core/srl/sft/__init__.py                       |      |
| read    | beni/core/hub/templates/model_card.md.j2            |      |
| process | pid 1077702                                         |      |

**Run 4: sft.yaml**

| # | Snapshot   | Time     | Root             | Changed |
|---|------------|----------|------------------|---------|
| 0 | run_start  | 06:19:31 | 32cfabf37bf320ae | 135     |
| 1 | phase eval | 06:19:32 | 32cfabf37bf320ae | 0       |
| 2 | run_end    | 06:21:41 | 32cfabf37bf320ae | 0       |

| Event | Path or command                           | Hash |
|-------|-------------------------------------------|------|
| read  | beni/__init__.py                          |      |
| read  | beni/cli/main.py                          |      |
| read  | beni/core/language.py                     |      |
| read  | beni/utils/__init__.py                    |      |
| read  | beni/utils/res/mdf_fields.json            |      |
| read  | beni/utils/res/language_metadata.json     |      |
| read  | beni/utils/res/iso_code_639_3.json        |      |
| read  | beni/core/srl/__init__.py                 |      |
| read  | beni/core/srl/config.py                   |      |
| read  | beni/core/safety/__init__.py              |      |
| read  | beni/core/safety/governor.py              |      |
| read  | .github/workflows/docs.yml                |      |
| read  | __init__.py                               |      |
| read  | beni/cli/__init__.py                      |      |
| read  | beni/core/__init__.py                     |      |
| read  | beni/core/compute/__init__.py             |      |
| read  | beni/core/compute/helpers.py              |      |
| read  | beni/core/compute/metrics.py              |      |
| read  | beni/core/compute/rewards.py              |      |
| read  | beni/core/hub/__init__.py                 |      |
| read  | beni/core/hub/model_card.py               |      |
| read  | beni/core/morphotactic/__init__.py        |      |
| read  | beni/core/morphotactic/distil/__init__.py |      |

| Event | Path or command                                          | Hash |
|-------|----------------------------------------------------------|------|
| read  | beni/core/morphotactic/distil/distillation.py            |      |
| read  | beni/core/morphotactic/distil/providers/__init__.py      |      |
| read  | beni/core/morphotactic/distil/providers/gguf.py          |      |
| read  | beni/core/morphotactic/distil/providers/google.py        |      |
| read  | beni/core/morphotactic/distil/providers/openai_compat.py |      |
| read  | beni/core/pipeline.py                                    |      |
| read  | beni/core/srl/algorithm1.py                              |      |
| read  | beni/core/srl/dpo/dpo.py                                 |      |
| read  | beni/core/srl/grpo/__init__.py                           |      |
| read  | beni/core/srl/grpo/callbacks.py                          |      |
| read  | beni/core/srl/jax/__init__.py                            |      |
| read  | beni/core/srl/jax/policy.py                              |      |
| read  | beni/core/srl/sft/sft.py                                 |      |
| read  | beni/core/srl/unified.py                                 |      |
| read  | beni/core/wordfreq/__init__.py                           |      |
| read  | beni/data/baselines/bmq/ex.json                          |      |
| read  | beni/data/datasets.py                                    |      |
| read  | beni/data/exp.yaml                                       |      |
| read  | beni/data/test.json                                      |      |
| read  | beni/utils/config.py                                     |      |
| read  | beni/utils/hf_lang.py                                    |      |
| read  | beni/utils/res/__init__.py                               |      |
| read  | cleanup.sh                                               |      |
| read  | configs/apo_bam.yaml                                     |      |
| read  | configs/dpo_bam.yaml                                     |      |
| read  | configs/exp.yaml                                         |      |
| read  | configs/grpo_bam.yaml                                    |      |
| read  | configs/grpo_multilang.yaml                              |      |
| read  | configs/matrix/base_multi13.yaml                         |      |
| read  | configs/matrix/models.json                               |      |
| read  | configs/presets/multi13.yaml                             |      |
| read  | configs/presets/single.yaml                              |      |
| read  | conftest.py                                              |      |
| read  | cookbooks/eval_apo.ipynb                                 |      |
| read  | cookbooks/eval_dpo.ipynb                                 |      |
| read  | cookbooks/eval_grpo.ipynb                                |      |
| read  | cookbooks/eval_sft.ipynb                                 |      |
| read  | cookbooks/experimentation_matrix.ipynb                   |      |
| read  | cookbooks/kveritas/sft.yaml                              |      |
| read  | cookbooks/kveritas_arm.py                                |      |
| read  | cookbooks/run_exp.ipynb                                  |      |
| read  | docs/javascripts/mathjax.js                              |      |
| read  | docs/javascripts/typewriter.js                           |      |
| read  | experiments/matrix/gemma-3-270m_apo/config.yaml          |      |
| read  | experiments/matrix/gemma-3-270m_dpo/config.yaml          |      |
| read  | experiments/matrix/gemma-3-270m_grpo/config.yaml         |      |
| read  | experiments/matrix/gemma-3-270m_sft/config.yaml          |      |
| read  | experiments/matrix/gemma-4-e2b_apo/config.yaml           |      |
| read  | experiments/matrix/gemma-4-e2b_dpo/config.yaml           |      |
| read  | experiments/matrix/gemma-4-e2b_grpo/config.yaml          |      |

| Event | Path or command                                           | Hash |
|-------|-----------------------------------------------------------|------|
| read  | experiments/matrix/gemma-4-e2b_sft/config.yaml            |      |
| read  | experiments/matrix/gemma-4-e4b_apo/config.yaml            |      |
| read  | experiments/matrix/gemma-4-e4b_dpo/config.yaml            |      |
| read  | experiments/matrix/gemma-4-e4b_grpo/config.yaml           |      |
| read  | experiments/matrix/gemma-4-e4b_sft/config.yaml            |      |
| read  | experiments/matrix/granite-4.0-h-350m_apo/config.yaml     |      |
| read  | experiments/matrix/granite-4.0-h-350m_dpo/config.yaml     |      |
| read  | experiments/matrix/granite-4.0-h-350m_grpo/config.yaml    |      |
| read  | experiments/matrix/granite-4.0-h-350m_sft/config.yaml     |      |
| read  | experiments/matrix/lfm2.5-230m_apo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-230m_dpo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-230m_grpo/config.yaml           |      |
| read  | experiments/matrix/lfm2.5-230m_sft/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-350m_apo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-350m_dpo/config.yaml            |      |
| read  | experiments/matrix/lfm2.5-350m_grpo/config.yaml           |      |
| read  | experiments/matrix/lfm2.5-350m_sft/config.yaml            |      |
| read  | experiments/matrix/matrix_results.json                    |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_apo/config.yaml  |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_dpo/config.yaml  |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_grpo/config.yaml |      |
| read  | experiments/matrix/phi-3.5-mini-instruct_sft/config.yaml  |      |
| read  | experiments/matrix/qwen2.5-0.5b_apo/config.yaml           |      |
| read  | experiments/matrix/qwen2.5-0.5b_dpo/config.yaml           |      |
| read  | experiments/matrix/qwen2.5-0.5b_grpo/config.yaml          |      |
| read  | experiments/matrix/qwen2.5-0.5b_sft/config.yaml           |      |
| read  | experiments/matrix/qwen3-0.6b_apo/config.yaml             |      |
| read  | experiments/matrix/qwen3-0.6b_dpo/config.yaml             |      |
| read  | experiments/matrix/qwen3-0.6b_grpo/config.yaml            |      |
| read  | experiments/matrix/qwen3-0.6b_sft/config.yaml             |      |
| read  | experiments/matrix/smollm2-135m_apo/config.yaml           |      |
| read  | experiments/matrix/smollm2-135m_dpo/config.yaml           |      |
| read  | experiments/matrix/smollm2-135m_grpo/config.yaml          |      |
| read  | experiments/matrix/smollm2-135m_sft/config.yaml           |      |
| read  | experiments/matrix/smollm3-3b_apo/config.yaml             |      |
| read  | experiments/matrix/smollm3-3b_dpo/config.yaml             |      |
| read  | experiments/matrix/smollm3-3b_grpo/config.yaml            |      |
| read  | experiments/matrix/smollm3-3b_sft/config.yaml             |      |
| read  | experiments/matrix/tinyllama-1.1b_apo/config.yaml         |      |
| read  | experiments/matrix/tinyllama-1.1b_dpo/config.yaml         |      |
| read  | experiments/matrix/tinyllama-1.1b_grpo/config.yaml        |      |
| read  | experiments/matrix/tinyllama-1.1b_sft/config.yaml         |      |
| read  | mkdocs.yml                                                |      |
| read  | pyproject.toml                                            |      |
| read  | scripts/run_matrix.py                                     |      |
| read  | setup.cfg                                                 |      |
| read  | setup.py                                                  |      |
| read  | tests/__init__.py                                         |      |
| read  | tests/test_framework.py                                   |      |
| read  | tests/test_grpo_pipeline.py                               |      |

| Event | Path or command                                 | Hash |
|-------|-------------------------------------------------|------|
| read  | tests/test_kveritas_callback.py                 |      |
| read  | beni/core/safety/spec.py                        |      |
| read  | beni/core/srl/grpo/grpo.py                      |      |
| read  | beni/core/srl/plugin.py                         |      |
| read  | beni/core/compute/uwec.py                       |      |
| read  | beni/core/morphotactic/dabax.py                 |      |
| read  | beni/data/__init__.py                           |      |
| read  | beni/core/morphotactic/distil/providers/base.py |      |
| read  | beni/core/srl/dpo/__init__.py                   |      |
| read  | beni/core/srl/apo/__init__.py                   |      |
| read  | beni/core/srl/apo/apo.py                        |      |
| read  | beni/core/srl/sft/__init__.py                   |      |
| read  | beni/data/baselines/mey/baseline.gram           |      |
| read  | beni/data/baselines/mey/baseline.dict           |      |
| read  | beni/data/baselines/kao/baseline.gram           |      |
| read  | beni/data/baselines/kao/baseline.dict           |      |

## C Run history

### Invocation 1, OK, 1.9 seconds, 6 output lines, started 2026-10-05T05:45:09

metric\_hash e3b0c44298fc1c149afbfc4c8996fb92427ae41e4649b934ca495991b7852b855  
run\_digest d5a8d3dd3a75b05752126f9afe5d6681d21b3d7e60f2e458b47b38ac13b4b4c0  
anchored at run end

### Invocation 2, OK, 55.1 seconds, 36 output lines, started 2026-10-05T05:45:11

metric\_hash e3b0c44298fc1c149afbfc4c8996fb92427ae41e4649b934ca495991b7852b855  
run\_digest 8eb7213262134e8855f0ec74157d5cbd7a8e60140a6d2ef6567f9b13907b35ee  
anchored at run end

### Invocation 3, OK, 33 minutes 22 seconds, 1103 output lines, started 2026-10-05T05:46:07

metric\_hash 5343e4ee7dd29eff1679bda48b226183d8e98cbceb084259b940f43b916903a7  
run\_digest ed5d804b6c21b073472bb7cea24a0063a910e3ca43f6c6803bee4150131fcf12  
anchored at run end

### Invocation 4, OK, 2 minutes 10 seconds, 70 output lines, started 2026-10-05T06:19:30

metric\_hash a30c845a667bb06b397e7909692da3f173e9108a9897806a21aad0715bf595c2  
run\_digest c5cedbac2a05872a4ddc5b17026eb23fd9a67b51bf6d39846feb945bff336842  
anchored at run end

## D Raw seal fields and hashes

### Seal

data\_hash 1c1699a5bb61643823134d4b20b8cbf76989b4df4327e1bd9ed91816f2e2d177  
signed\_message\_hash 23625794c2ef9a457f913296a2238de2fa5ad6e326a259a3fa9da3a81ad987cb  
nonce 2b81395b59db6c0dd5b58124d2acdd1b  
signed\_at 2026-10-05T06:21:42.666471+00:00  
sealed\_at 2026-10-05T06:21:42.72501772Z  
checkout\_bundle 199eb8b7c51638eae96fa0c03f3f5834f1b16c35debd164a5a4999b69388553a

### Signature (base64)

dDo/Cbx0mTzOXW779xClnfd0xu0lyTxmdGp1oGtUCqrhU/fAqS4TB/cyEedj6cgKqv40FQXMFh8LPEVXDhLPbj+7  
1uPCf8tDjXQJcJH02IaDvGuw0dDsn+m1Bfr+ltuo3acAQKMs0v56BCgbrQuUdF/38zu1YNQhqdphZ8mRa9f6VD  
BM5j5m44rXCzx104IRY9zVTZq/HZdsSGpm8VqQfMoBKxaoaZqVEsy08MLRJgpDwALy0RsLzu5pnt4VUr lwe0EvMT  
2m3T3hKzVi/ZGNDQ0a40+a6XFa5qMVsU6m/bQXJvLWp/c6b8EWnn3uQyHJhFPxY2UCSuHBCFtsjhUx6y0yFqLQ3h  
ymCUQFChvHWIKTwTmJLAQu130gBwmJLuzWjh9zB2Pa35wJthvnp1S19SvcHutqwaczaZRmNUOyZrM2KTIIfyVvyxZ

pNApynX88Q4uF2a6TEsvC7s70e8AySTslQpvKTFxyKE0nf7zQDkGvyPUD2QeoHU/5CMmuWwfokv0a2njEgH8A8Kf  
 lD+IHn9YIOGwdXhVT7oFjOuK6VzJc03APEzN6eBQ/mZCZYmiC4YPJ0Q6p3aTJK/yvXT+e+KX+rJxx4jsuL+La6Pi  
 UiFu67B6X8ze0NRX0RifF0b8GBRHmsgLNTpGbhl0G30vyIQSGTRVDAS/hUo0tIJHQDo=

### Public key

```
-----BEGIN PUBLIC KEY-----
MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEA1QL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcPrOPPCqp/LouFz5CUZzRi
6+8FTv70dHuIz5A4LAT4vcgwP+iWscPc3HzpmjpsKbTm0emqgp6RB6GNnPpxYa5X
7qU95EfEisX/GfoS9ToXtqPBwCcVWSPBX8efec4TC58m96Fsp4mC8KlCeadbIbHN
av0JjdTJ//CM0WZSIbdzE6qvEbNFcqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdHQ+D
AhfTRuJNgWEkjwD5lb0Vhi4qlkweSOEMTKT3ye9Il5xEaJn4bku4eKqes7VSZt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPgPZ242NH8JgE6CNF7ToL9syasS67u7UMgIcNgeNHNcy9bLgpU5annzP9gN2l
6HEMldkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtfo4R5UGVV92qd2Mi
1GN7lE3mbQRbPsLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/olCHBCf1VIDCKdxohehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PWjNcNku5cdw3gHlCCAwEAAQ==
-----END PUBLIC KEY-----
```

### Run 1 outputs

```
stdout 95da3d4ccdc2cb36634021493dd22ede9e9a03b3581cbffe53992258d08bf806
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
digest d5a8d3dd3a75b05752126f9afe5d6681d21b3d7e60f2e458b47b38ac13b4b4c0
cookbooks/kveritas_arm.py (unchanged)
pre c221e36799c1e2d1e6ea95a6fd46e514b4f99444da247695e531964b3b67c67d
post c221e36799c1e2d1e6ea95a6fd46e514b4f99444da247695e531964b3b67c67d
```

### Run 2 outputs

```
stdout d710f5a2fe5892fb5c2da8a226b7f6eb56e846d189411b82ea5cebe96aa899e6
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
digest 8eb7213262134e8855f0ec74157d5cbd7a8e60140a6d2ef6567f9b13907b35ee
```

### Run 3 outputs

```
stdout b7426cfead30b4bd16e16a120544196408d2458654c2e521ae87a2647ea0ab9f
stderr 6df27421dc4eba6292dda271ee5d2ac77d623882cdf6aef4af54e167d4ec132b
metrics 5343e4ee7dd29eff1679bda48b226183d8e98cbceb084259b940f43b916903a7
digest ed5d804b6c21b073472bb7cea24a0063a910e3ca43f6c6803bee4150131fcf12
```

### Run 4 outputs

```
stdout 7ea7a42ab6fc3bc320793fcecc1054dc04274e80ddf0319b46877bd6c00d528d
stderr 8afe9ef6b4143e4bd42a5f425988b2b37a1daba2eb4aa66dff6ca7c5cdfde2a4
metrics a30c845a667bb06b397e7909692da3f173e9108a9897806a21aad0715bf595c2
digest c5cedbac2a05872a4ddc5b17026eb23fd9a67b51bf6d39846feb945bff336842
```

## E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},
  author={Mamadou K. Keita and Christopher Homan},
  year={2026},
  eprint={2605.08586},
  archivePrefix={arXiv},
  primaryClass={cs.CR},
  url={https://arxiv.org/abs/2605.08586},
}
```